Online Shop Management System

Course Name: COM337 ADVANCED PROGRAMMING.

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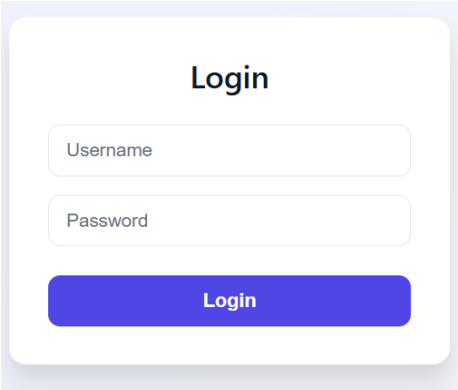
Introduction:

Visual elements make applications more engaging and user-friendly. After completing the first phase of the Online Shop Management System, which mainly focused on backend functionality, a frontend interface was added in this phase. The introduction of the frontend allows both users and administrators to effectively interact with the application, improving usability and overall system efficiency.

Frontend and Backend Connection (api.js)

The api.js file is used to connect the frontend with the backend of the online shop. It sends requests from the frontend to the backend and receives data in return. All API calls are written in one file to make the code easy to manage. This file handles login and logout, user management, product management, cart actions, payments, and admin data. It also keeps the user logged in by using sessions. Based on the response from the backend, the frontend shows messages and updates the page. The **api.js** file helps the frontend and backend work together smoothly.

Login page:



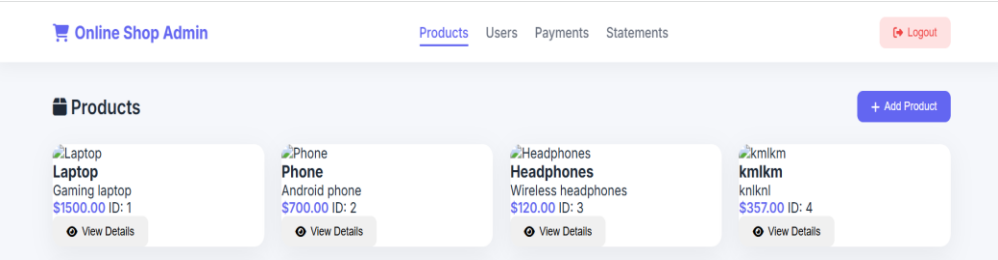

```java
@PostMapping("/login")
public String login(@RequestParam String username,
                    @RequestParam String password,
                    HttpServletRequest request) {


    var adminOpt = adminRepo.findByUsername(username);
    if (adminOpt.isPresent()) {
        Admin admin = adminOpt.get();
        if (passwordEncoder.matches(password, admin.getPassword())) {
                            HttpSession session = request.getSession();
            session.setAttribute("userType", "ADMIN");
            session.setAttribute("username", username);
            session.setAttribute("adminId", admin.getId());

            return "redirect:/admin.html";
    }
```

The system uses one login page for both users and administrators. After login, the backend checks whether the account belongs to an admin or a user. Admins are redirected to **admin.html**, while users are redirected to **shop.html**. Login information is stored using sessions to control access. When the user logs out, the session ends to keep the system secure.

Admin.html



The admin page allows administrators to manage the online shop. From this page, admins can add and update products, manage users, view payment details, and check sales statements. This page is accessible only to administrators after a successful login, ensuring secure control over system operations.

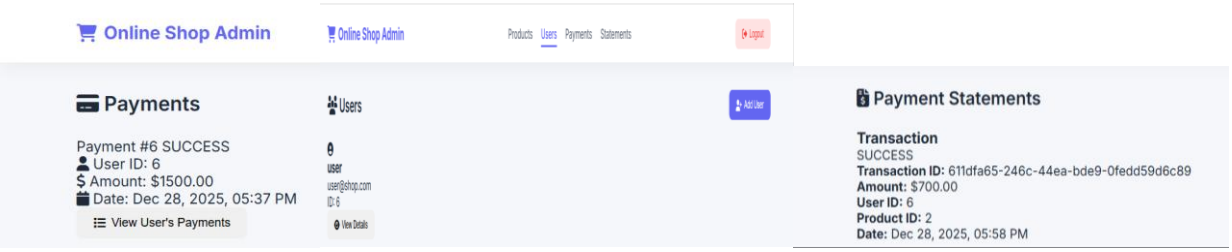
Admin Page Script (script.js)

The **script.js** file controls how the admin page works. It handles user actions, page navigation, and communication with the backend through the **api.js** file.

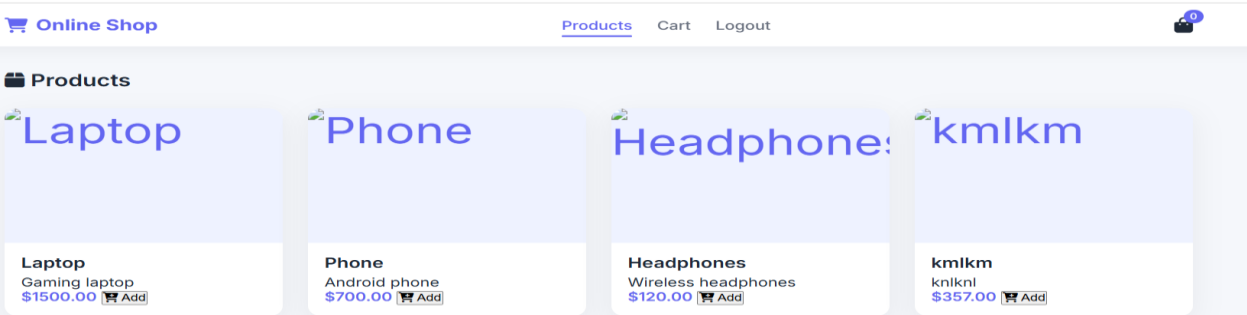
This script allows the administrator to:

- Switch between sections like products, users, payments, and statements.
- View lists of users, products, payments, and statements.
- Add new users and products using forms.
- View detailed information in pop-up windows (modals).
- See loading indicators while data is being fetched.
- Log out securely from the system.

The script updates the page dynamically without reloading it. It shows success or error messages based on the backend response. This file helps the admin manage the online shop easily through the frontend.



Shop.html



The **shop.html** page is the main dashboard for users after login. It allows users to view available products, manage their shopping cart, and make purchases. The page is connected to the backend using JavaScript files **(api.js** and **script2.js)** to load product data, update the cart, and handle user actions without reloading the page. A loading spinner is shown while data is being fetched to improve user experience.

Admin Script (script.js)

script.js handles the admin dashboard. It lets the admin manage users, products, payments, and statements by connecting the frontend to the backend.

User Shopping Script (script2.js)

The **script2.js** file controls how users shop in the online store. It loads products from the backend, shows them on the page, and allows users to add products to the cart. This script manages the shopping cart by increasing or decreasing item quantity, removing items, and showing the total price. Cart data is saved so it is not lost when the page refreshes. It also handles checkout, logout, and page navigation between products and cart sections. The page updates automatically without reloading, making shopping easy and smooth for the user.

Terminology Used in the Project

- **Frontend**: The part of the system that the user can see and interact with (HTML, CSS, JavaScript).
- **Backend**: The server-side part that handles logic, database, and security (Spring Boot).
- **API (Application Programming Interface)**: A bridge that allows the frontend and backend to communicate.
- **Controller**: A backend class that receives requests from the frontend and sends responses.
- **Repository**: Used to interact with the database (save, update, delete, fetch data).
- **Entity**: A Java class that represents database tables such as User, Admin, Product, and Payment.
- **HTTP Request**: A message sent from frontend to backend (GET, POST).
- **HTTP Response**: Data sent back from backend to frontend.
- **Session**: Stores user login information during browsing.
- **Authentication**: Checking if the user or admin is logged in.
- **Authorization**: Deciding what actions a user or admin is allowed to perform.
- **JSON**: Data format used to send information between frontend and backend